

Shrinking Quantum Hall Edge Simulator

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Abstract

Expanding Quantum Hall systems hold promise for their ability to replicate the expansion experienced in various cosmological phenomena: black holes, general relativity, and the expansion of the universe. Fabricating chips to run expanding Quantum Hall edge experiments is time-intensive work. This paper reports my work on a Python command-line application to simulate a user-provided chip during an experiment. This simulation is able to compute the effects of applying a gate voltage to an expanding region as electrons are traveling along it. It computes various metrics of performance for the chip, such as scale factor and edge width, and produces several graphs that give insight into problems that may occur for a given configuration. The configurations on which I ran the simulation highlight the need for a smooth material edge and circular symmetry, as well as isolating the scale factor frequency shift from the velocity frequency shift. This simulation can inform experimental design for expanding Quantum Hall edge systems and give insight into interpreting experimental results to enable more precise and quicker advancement of Quantum Hall technology.

1 Introduction

Background

According to the theory of general relativity, space and time curve in the presence of mass, represented by the stress-energy tensor. This gives rise to many interesting and confusing cosmological and quantum phenomena, such as a slight redshift, Hawking radiation, and the Hubble horizon.

We introduce an analog curvature in the Quantum Hall edge on a GaAs chip. A Quantum Hall edge is a 1+1-dimensional (1 space dimension, 1 time dimension) quantum system. It is described by a bulk region in which the electrons cannot move due to being confined by a magnetic field, and a 1-dimensional edge in which the electrons can move. Our experiment involves changing the shape of that edge as the electrons move along it. By making it longer, we are effectively curving the space that the electrons travel along, creating an analog to space curvature that is experienced due to general relativity.

My Work

My work involves changing the shape of the edge on the GaAs chip and keeping the edge well-defined. The important metric in question is the scale factor $a(t)$, which describes the relative length of the edge, and thus how much space has been effectively stretched. The question is how to produce the $\tilde{a}(t)$ that we want.

Importance

Answering this question and creating a model with a well-defined edge will let us make a Quantum Hall system that can measure Hawking radiation and experimentally determine what happens at wavelengths beyond the Hubble horizon. We can also observe quantum particle creation. This could help answer questions about what happens inside a black hole and what happened in the early stages of the still-expanding universe.

This type of expanding-edge Quantum Hall system has also been shown to facilitate long-range quantum energy teleportation (QET). The normal formulations of QET show that the amount of energy that can be transported over a distance L scales with $O(L^{-1})$. However, performing QET with an expanding edge allows the energy to travel arbitrarily far; that is, the amount of energy that can be transported over a distance L scales with $O(L^0)$. Thus, expanding-edge Quantum Hall systems can make practical QET possible.

Impact

Humans have always been curious about the structure of the universe. What child isn't fascinated by the concept of a black hole, and what philosopher is uninterested in where it all came from? My work will help give a fuller description by looking at some of the most confusing concepts in cosmology and quantum field theory. My work also helps make possible technological advancements in energy transportation via QET.

2 Background and Setup

Our experiment involves a GaAs chip shaped as the yellow region in Figure 1. We cool the chip down to a low temperature and apply a powerful magnetic field perpendicular to the chip. We then generate a pulse of electrons along the Quantum Hall edge and apply an increasing gate voltage to the chip as the pulse is traveling along it.

To simulate the chip, I implemented a Python command-line interface program. It allows the user to input a configuration file with the following:

- The size and shape of the chip as an array of points with boolean values (1 for in the chip, 0 for not in the chip). Each point represents a pixel of size $20l_b$ where l_b is the magnetic length of the system.
- The places to apply a gate voltage.
- The configuration gate potential as an array of potential values.
- The time between values in the configuration gate potential.
- A desired scale factor as an array of values.
- A function that converts applied gate voltage to effective potential.
- The starting position of an electron along the edge to simulate.

The command-line interface then imports the configuration file and produces the figures shown (as well as a couple more).

Quantum Hall effect

The Quantum Hall effect occurs in a semiconductor at low temperatures in the presence of a powerful magnetic field. When the Landau level of the material approaches an integer or rational fraction, the electrons in the material will be confined to travel in small circles (as electrons are wont to do in the presence of a magnetic field). However, at the edge of the material, electrons cannot complete a full circle. They bounce and travel along the edge. Therefore, a thin conducting edge is created next to the insulating layer of the bulk.

Edge Definition

The edge is not precisely at the material edge: a more prudent definition is where electrons have energy roughly equal to the Fermi energy of the Landau level (for our experiment, we use the lowest Landau level). We model this by defining a fluctuation energy calculated using the temperature of the system and its disorder and define the edge as any points where the energy is within the fluctuation energy of the Fermi energy:

$$U(\vec{x}) - U_{fluc} < E_F < U(x) + U_{fluc}$$

Energy Structure

To calculate the position of the edge, we compute the energy structure of the material. Electrons cannot exit the GaAs, which we model as a standard inverse square potential:

$$U(\vec{x}) \sim \frac{A}{x^2}$$

We tuned the parameter A to fit with our experimental findings of the distance from the Quantum Hall edge to the material edge.

We additionally apply a minimum ground state energy to the material $E_G = E_F/2$, resulting in the energy structure seen in Figure 2. The location of the resulting Quantum Hall edge is shown in Figure 3.

Gate Potential

To model the increasing gate voltage, we add a small amount of energy, incremented over 100 time steps, to the material around the shrinking region. The energy structure at time step 50 is shown in Figure 4.

3 Results

Scale Factor

To compute the length of the edge at each time step, I first make the assumption that the edge at each y coordinate is where $U(x) = E_F$ (computed precisely by using a 4th degree polynomial interpolation around the neighboring points above and below the Fermi energy) while checking the the edge is not too wide so that this is not a significant deviation from any path the electron would take. Once the computed edge is truly 1-dimensional, I simply sum the lengths to the next y coordinate in order to calculate the edge length. Dividing by the initial edge length gets us the scale factor shown in Figure 5.

In order to generate scale factors analogous to those for a black hole or an expanding universe, my simulation allows the user to input a desired scale factor. After computing the edge length at each time step in the configuration gate potential, it computes and returns the potential that produces the closest edge length to the desired edge length. The results of this algorithm are shown in Figure 6, where the desired scale factor is $\cosh^2(t)$ (scaled appropriately), the same scale factor that is experienced in an expanding De Sitter universe. This algorithm was similarly successful for many other configurations that I simulated.

Dynamics

The velocity of an electron traveling along a Quantum Hall edge is given by the formula:

$$v = \frac{|\nabla U(\vec{x})|}{B \cdot e}$$

Here, B is the magnitude of the perpendicular magnetic field and e is the elementary charge. The results of this formula are shown at time steps 0 and 50 in Figure 2 and Figure 4 respectively. The noise in the center region is presumably due to poor pixelation; the distance from the material edge is not smooth for a pixelated circular material edge. For a truly circular material edge, the velocity in the expanding region should be uniform.

With the velocity along the edge at each time step and edge position at each time step, I computed the path of the electron numerically under the assumption that the electron retains its y -coordinate when the edge moves. The results are shown in Figure 7.

I also computed the local expansion at each time step; that is, the ratio of the length of a small neighborhood of the edge at one point to the length of that neighborhood at the next time step. Since the edge is an equipotential region, it is perpendicular to the gradient of the energy and thus perpendicular to the direction of motion of the edge as a small amount of energy is applied. Therefore, the position of the edge at each time step also gives the direction of its motion, and thus the projection of a small region of edge surrounding a point onto the edge at the next time step can be calculated (with vast amounts of trigonometry). The time-averaged (geometric mean) expansion ratio at each y -coordinate is shown in Figure 8. The result is very noisy due to poor pixelation. Considering that in the expanding region the edge remains circular, the expansion should be uniform. The result, however, is too noisy to tell.

Knowing the path the electron takes and the relative expansion at each point in space, I computed the relative expansion experienced by an electron traveling along the edge. Instead of producing the relative expansion at each time step, I chose to use a logarithm transform to compute the relative expansion more accurately. Defining $s(t)$ to be the relative expansion at each point in time for the electron and t_0 to be the number of time steps, we have the following:

$$\ln\left(\prod_{k=1}^N s\left(\frac{kt_0}{N}\right)^{\frac{t_0}{N}}\right) = \sum_{k=1}^N \frac{t_0}{N} \ln\left(s\left(\frac{kt_0}{N}\right)\right) \rightarrow \int_0^{t_0} \ln(s(t)) dt$$

At $N = t_0$, this is equivalent to producing the relative expansion at each time step; however, computing it as an integral gives more precision. Computing this integral for each time step as t_0 yields the relative expansion of the electron at each time step, and is shown in Figure 9. The result is very noisy and shows some expansion of the electron, something that shouldn't happen for the theoretically uniformly shrinking edge. It does, however, resemble the scale factor calculated earlier. I use the scale factor calculated earlier instead of this relative expansion due to the noise.

Pulse

As a preliminary experiment, we wish to verify the edge expansion and measure redshift. To do so, we create an electron pulse with frequency f_0 along the Quantum Hall edge. As the pulse is traveling along the edge, we apply a gate voltage to shrink the edge. We then measure the frequency f_1 of the pulse on the other side of the expanding region.

I simulated this, making sure that the pulse stayed fully in the expanding region for the full duration of the gate voltage by calculating the paths of the start and end of the pulse. I then computed the Fourier Transform of the pulse without applying a gate voltage and compared it against the Fourier Transform of the pulse when a gate voltage is applied.

An interesting result that I found was that the velocity affects the frequency. Naturally, uniformly shrinking the region that the pulse is in decreases the distance between nodes and leads to a blueshift. However, changing the position of the edge changes the gradient of the energy and therefore also changes the velocity. A comparison between the velocities with no gate voltage and some gate voltage is given in Figure 10. Decreasing the velocity as the pulse travels along the edge increases the time it takes for one wave node to get to the next and thus leads to a redshift. For this shift, I used the average velocity along the expanding region since theory says that it should be uniform. I computed the shift in frequency due to scale factor and velocity change separately and in combination. The results are shown in Figure 11.

Notably, a change in the velocity of electrons along the edge leads to a much larger shift than the scale factor. Scale factor blueshift is still measurable. Therefore, this experiment should be able to detect a blueshift in the electron pulse due to the expansion of space. Running the gate voltage in reverse, this experiment should be able to measure a redshift due to the expansion of space.

4 Discussion and Conclusion

I tested a position-dependent gate voltage constructed by depositing a varying thickness dielectric between the metal gate and the surface of the GaAs chip. This configuration was able to achieve a scale factor as low as .82 and retained a relatively low edge width. It was also able to achieve much lower noise in the velocity and expansion due to the larger distance from the material edge. The expansion, however, was non-uniform due to the circular asymmetry of the deposited dielectric.

I also tested a configuration with several semicircular steps added to the potential. This configuration was able to achieve a scale factor of .84, however, the edge width got moderately high ($\sim 4\mu m$). The expansion and

velocity also spiked several times as each step passed the Fermi energy when applying gate voltage, leading to irregularities computed in the dynamics.

My simulation proved very useful in computing various quantities of the Quantum Hall device and gave insights as to the performance of various configurations of the experiment. It also proved very versatile in dealing with different configurations. I believe it can be well-adapted to suit other experiments one may do in such an expanding Quantum Hall system. The part to compute the Fourier Transforms of an electron pulse was quite simple with the rest of the framework provided by my simulation; I believe adding another section to suit another experiment would be equally simple and very valuable.

Unexpectedly, the pixelation had a large effect on computing the velocity and relative local expansion. A more improved version could use a better resolution to avoid this issue. The pixelation also highlights how important it is to have a smooth material edge when fabricating. Also unexpectedly, the velocity changes when applying gate voltage, leading to a frequency shift that opposes the scale factor shift. The scale factor shift is still theoretically measurable, but if the material edge is not smooth and the velocity is not uniform, there may be too much noise. This again highlights the need for a smooth material edge and precise, low-noise conditions.

Future work on the simulation would include a greater spatial resolution to avoid pixelation-related issues, with a resolution up to the resolution of the material edge. A greater time resolution should help with the dynamics and give better insights into the expansion an electron experiences along the edge. Additionally, a more precise definition of the edge would help, as the electrons' thermal energy ultimately follows a Gaussian distribution, and some electrons outside $E_F \pm U_{fluc}$ have enough energy to move. Based on that, an average path could be calculated, which is different from the center of the edge that I computed. A more precise parameter for the confinement potential could be implemented to ensure the simulation is accurate. I simulated very angular steps for the configuration, where the actual manufacturing technique would produce more rounded steps. Replacing that in the configuration should make the results smoother and more regular. This simulation also invites other experiments with expanding Quantum Hall edges, such as measuring the Hubble constant and Hawking radiation.

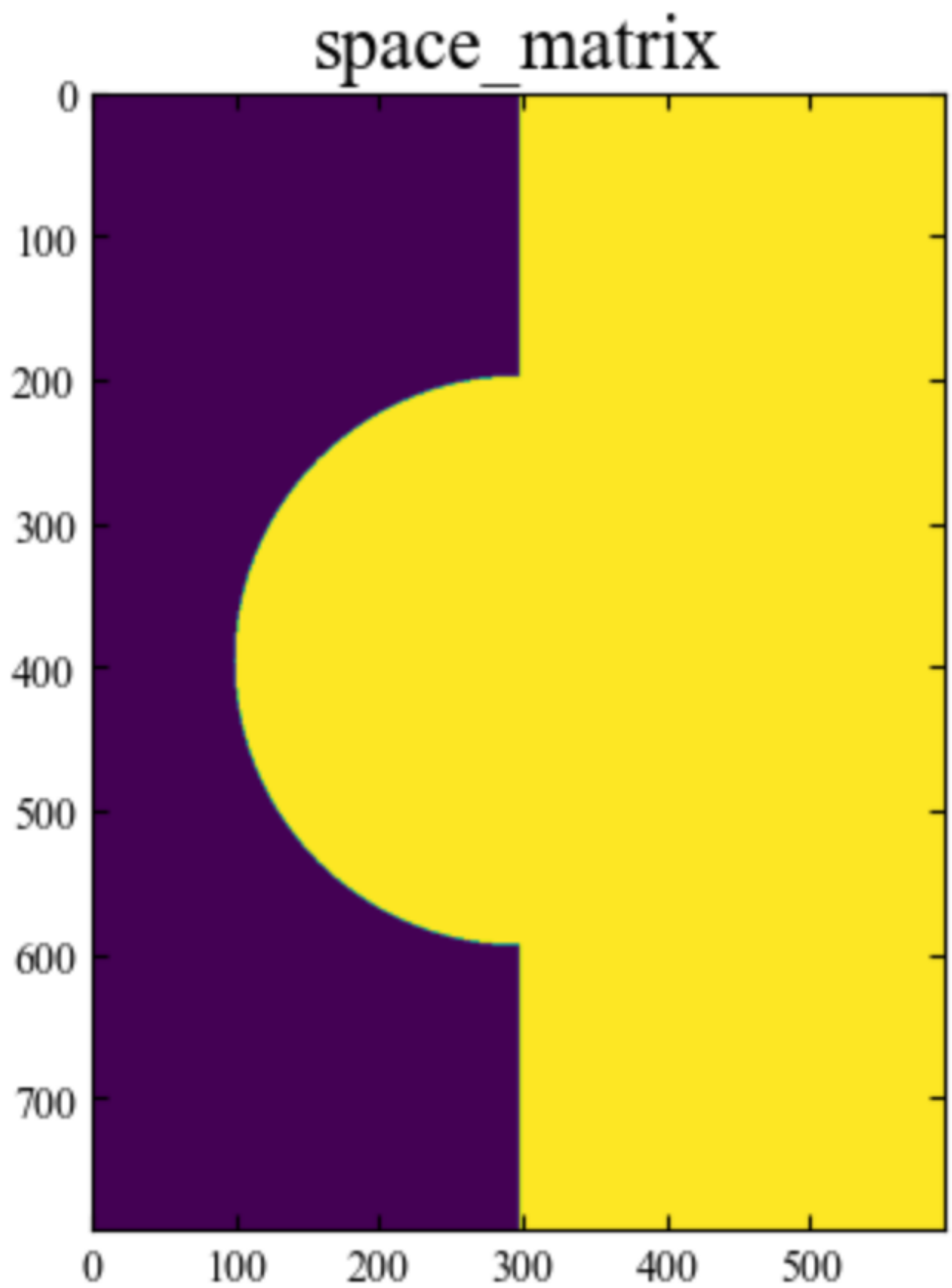


Figure 1: Quantum Hall effect

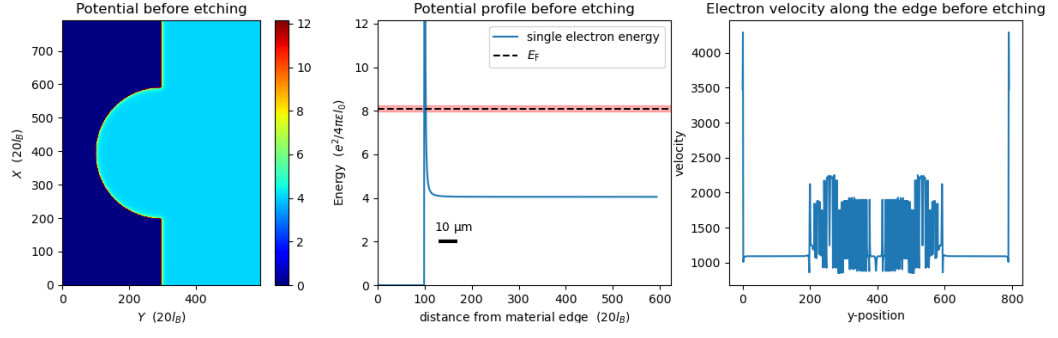


Figure 2: Energy structure without gate voltage

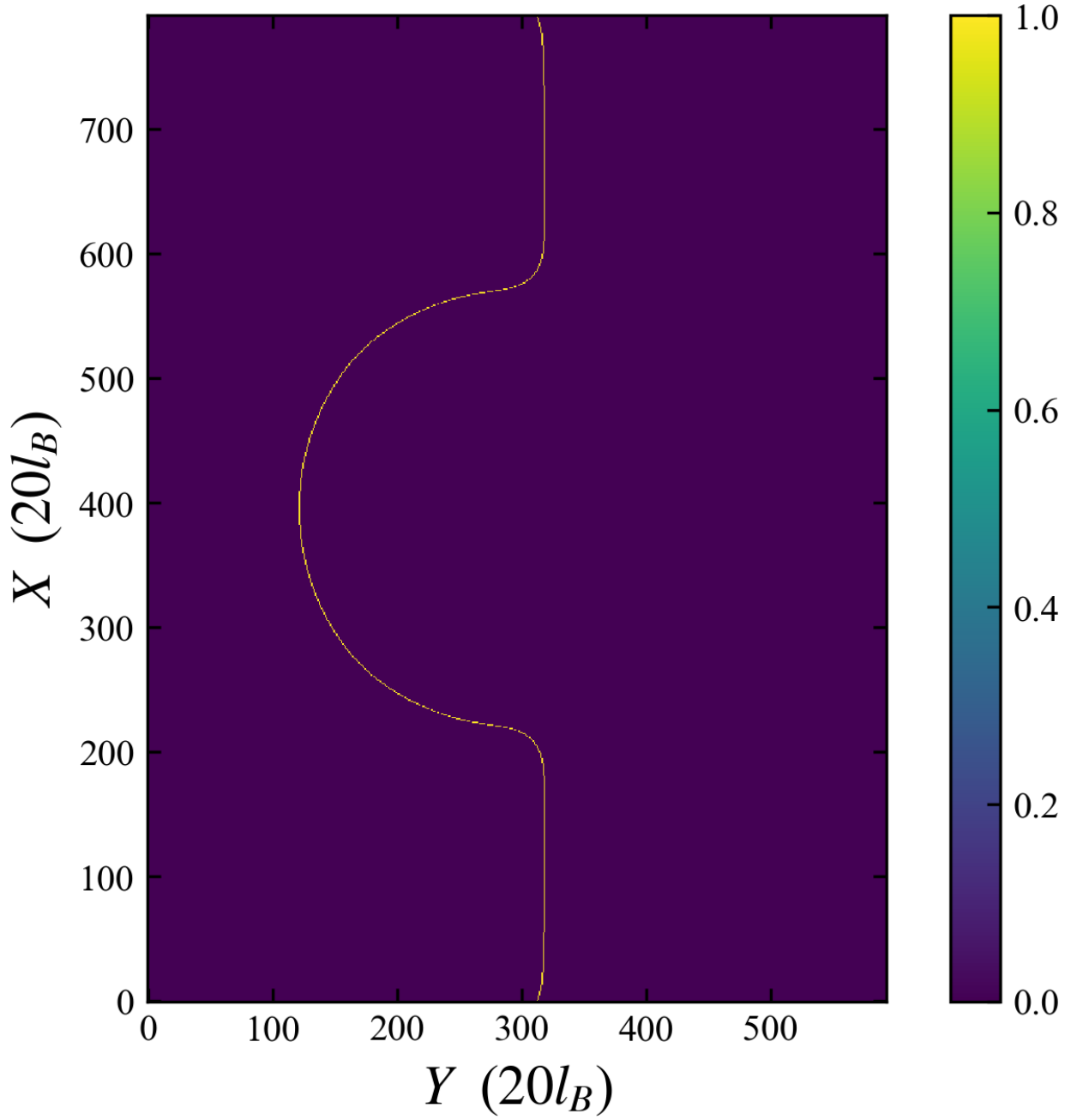


Figure 3: Edge position

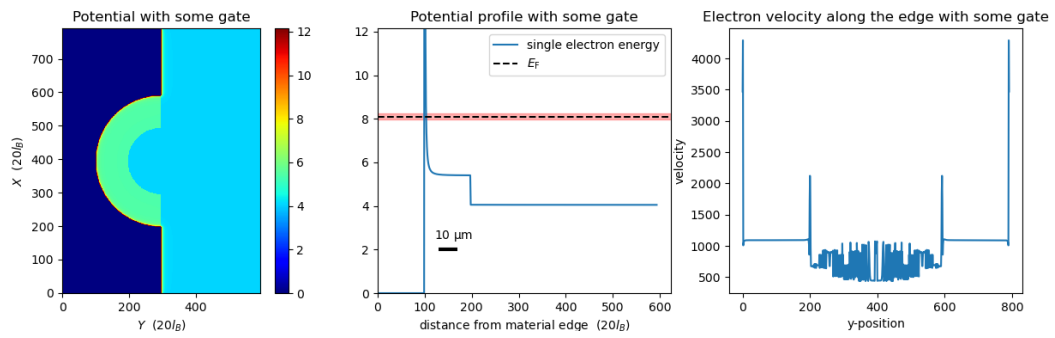


Figure 4: Energy structure with some gate voltage

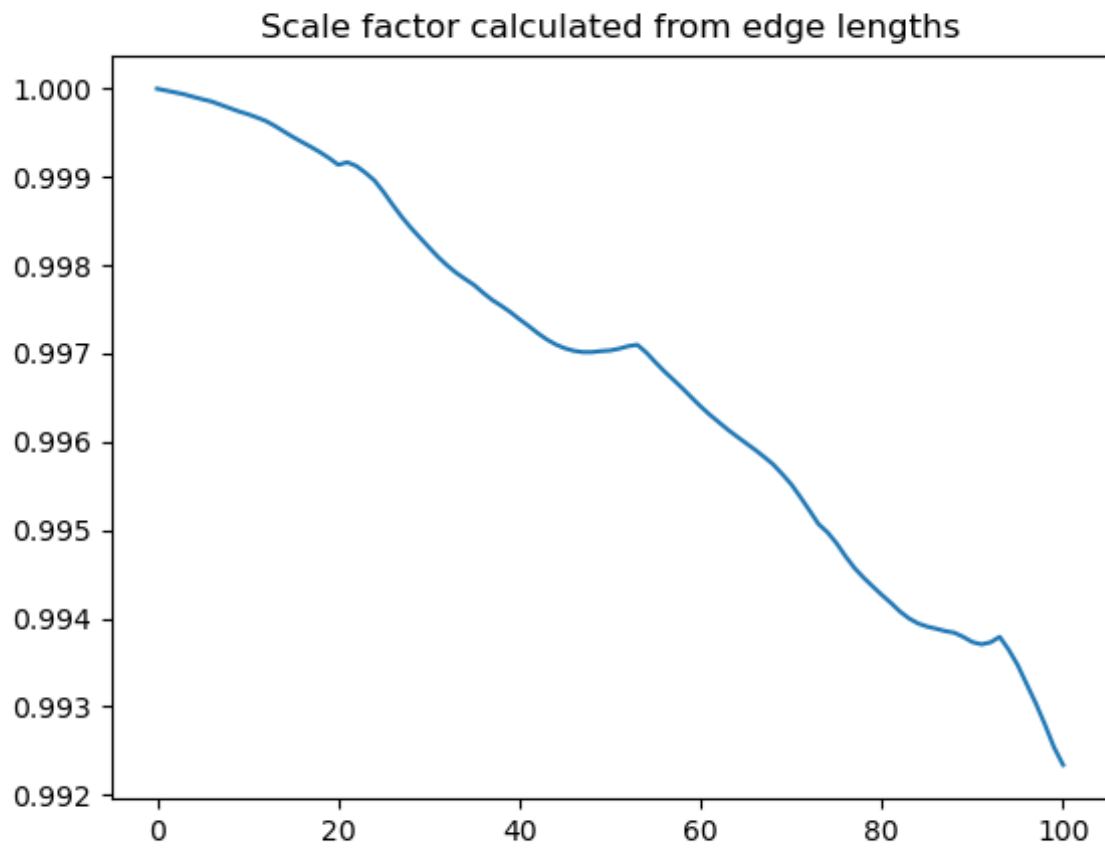


Figure 5: Scale factor

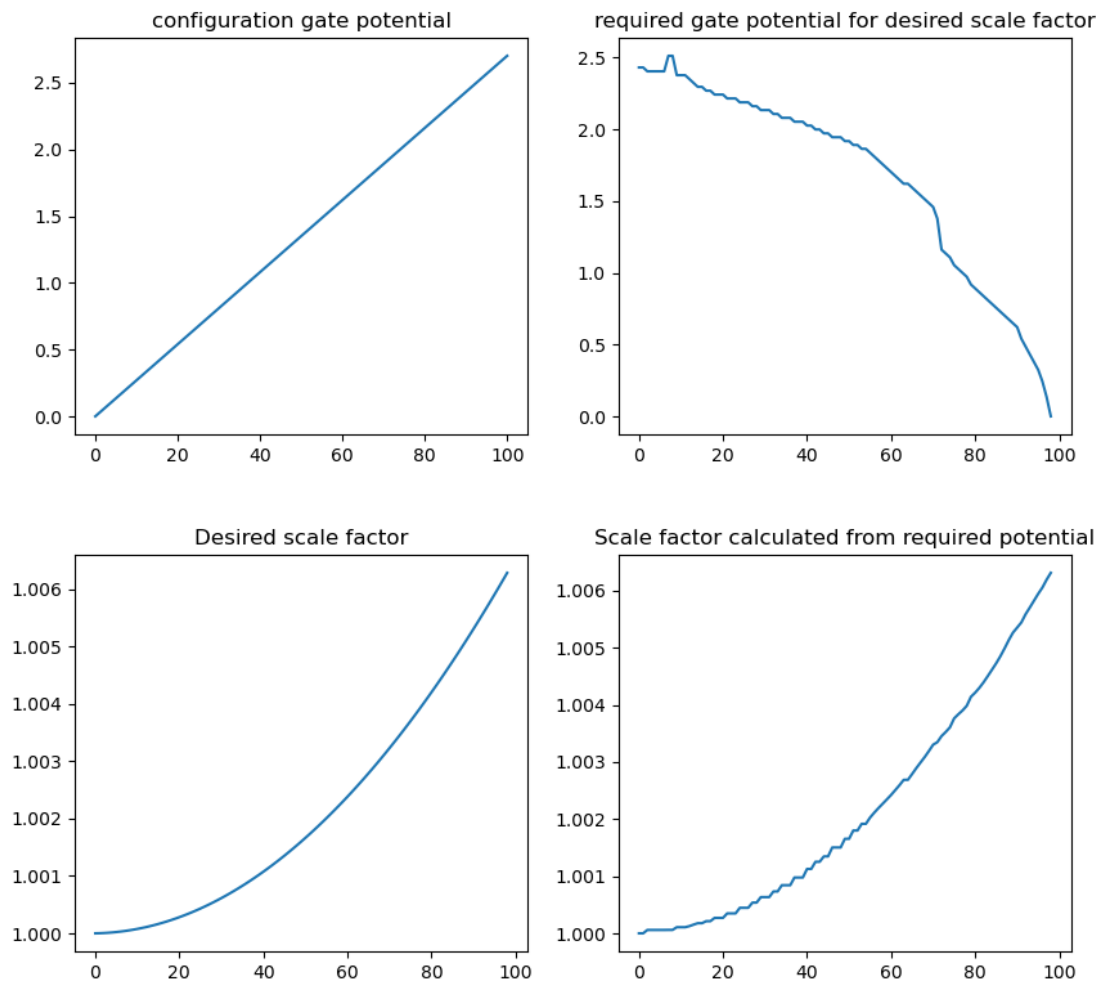


Figure 6: Scale factor algorithm

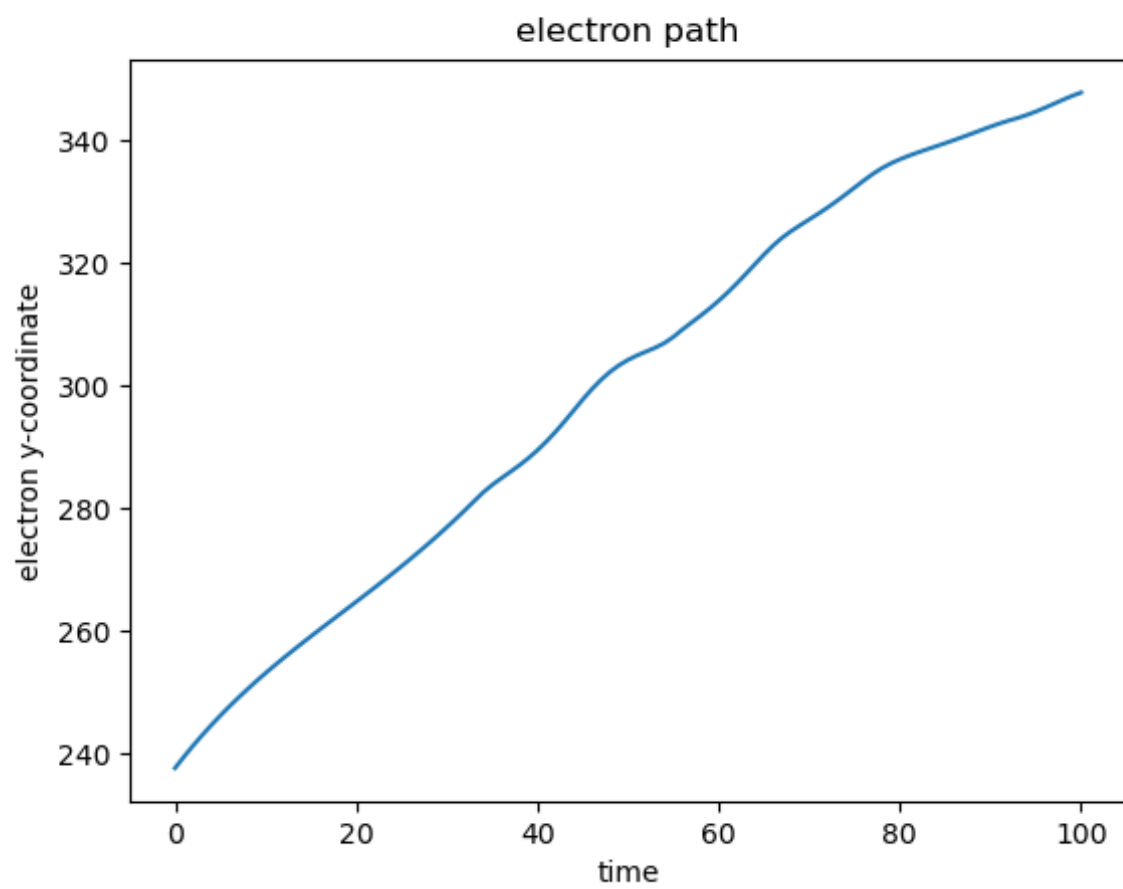


Figure 7: Electron path

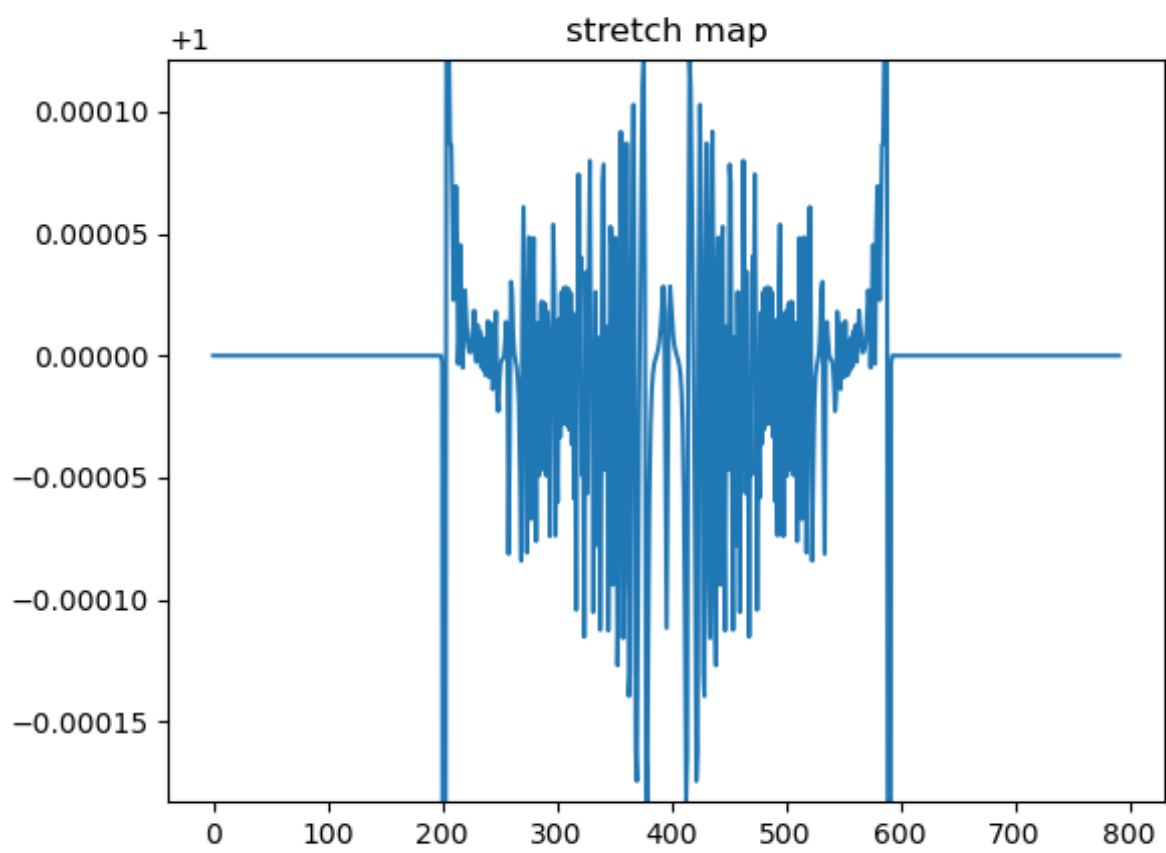


Figure 8: Local average expansion ratio

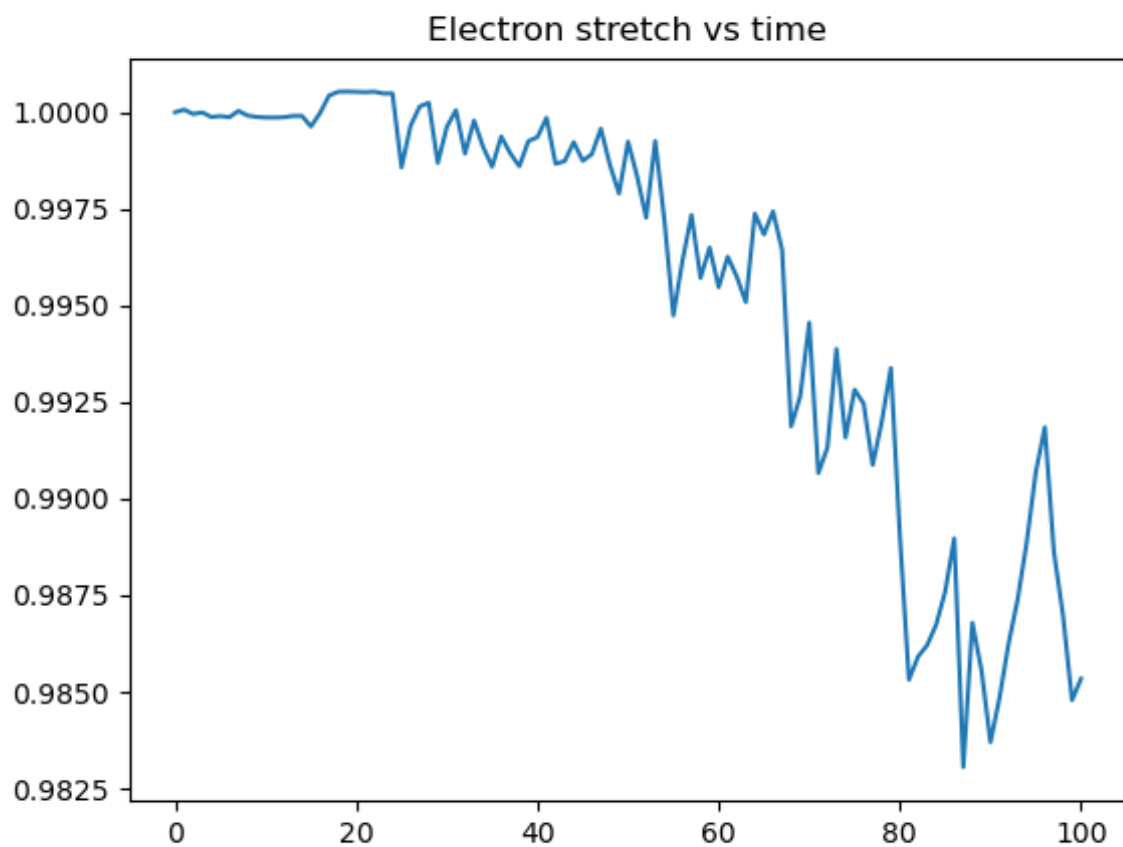


Figure 9: Dynamic relative expansion of electron

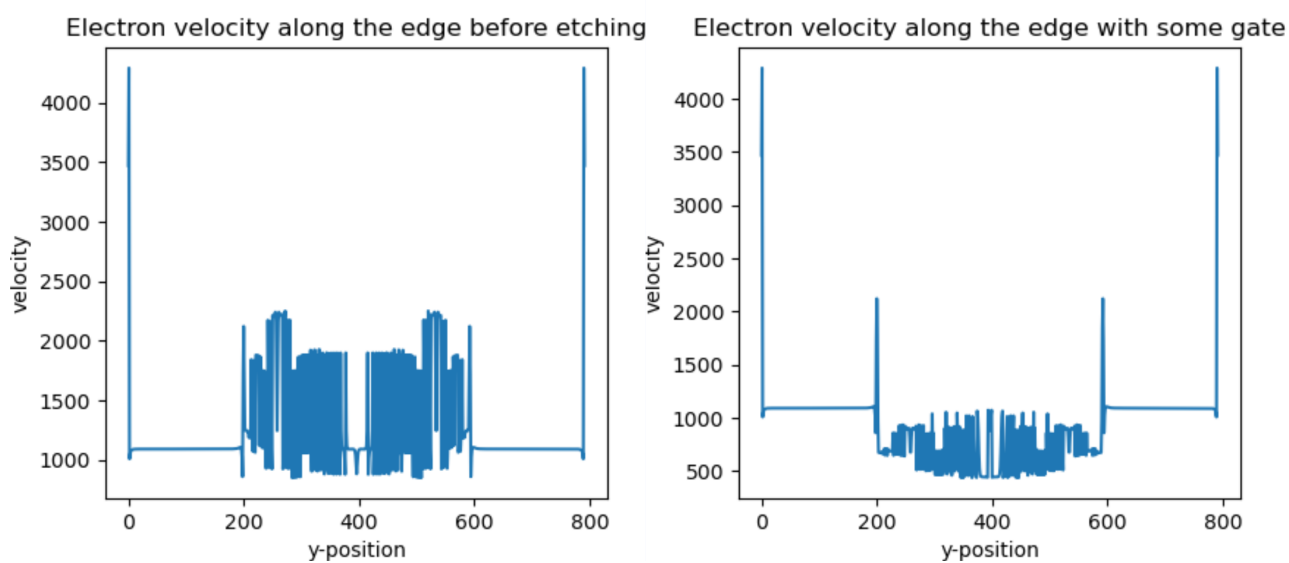


Figure 10: Velocity comparison - no gate voltage vs some gate voltage

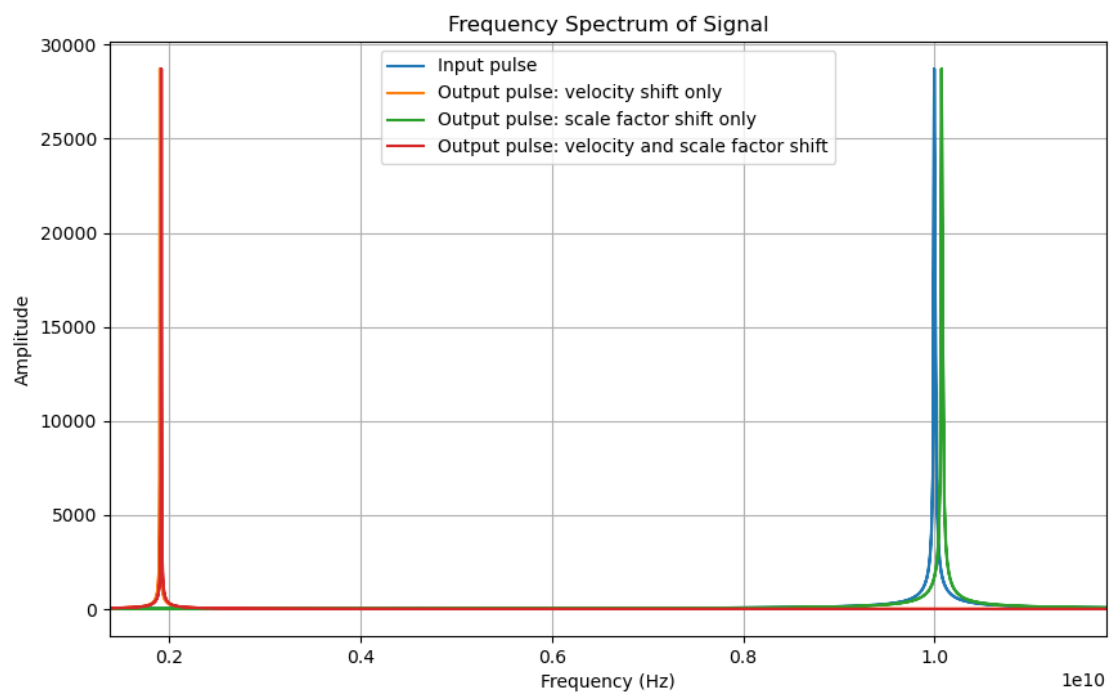


Figure 11: Fourier transforms of electron pulse